Purchasing Goods

Managing Money

2026

Preliminary Mathematics Standard

Outline

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Percentage Increase & Decrease

Percentage Increase:

$$\text{New Price} = \text{Original} \times \left(1 + \frac{r}{100}\right)$$

Percentage Decrease:

$$\text{New Price} = \text{Original} \times \left(1 - \frac{r}{100}\right)$$

Example: A jacket costs \$120 and is discounted by 15%.

$$120 \times (1 - 0.15) = 120 \times 0.85 = \$102$$

Profit and Loss

Profit occurs when selling price > cost price.
Loss occurs when selling price < cost price.

$$\text{Profit/Loss} = \text{Selling Price} - \text{Cost Price}$$

$$\text{Profit/Loss \%} = \frac{\text{Profit or Loss}}{\text{Cost Price}} \times 100$$

Example: A bike bought for \$350 is sold for \$420.

$$\text{Profit} = \$420 - \$350 = \$70$$

$$\text{Profit \%} = \frac{70}{350} \times 100 = 20\%$$

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Goods and Services Tax (GST)

In Australia: GST is a 10% tax added to most goods and services.

Adding GST:

$$\text{Price inc. GST} = \text{Pre-tax price} \times 1.10$$

Removing GST:

$$\text{Pre-tax price} = \frac{\text{Price inc. GST}}{1.10}$$

GST component only:

$$\text{GST amount} = \frac{\text{Price inc. GST}}{11}$$

Example: A pre-tax item costs \$85.

$$\$85 \times 1.10 = \$93.50$$

Value-Added Tax (VAT)

Internationally: VAT is similar to GST but rates vary by country.

Country	**Standard VAT Rate**
Australia (GST)	10%
United Kingdom	20%
Germany	19%
France	20%
New Zealand	15%

General formula (same structure as GST):

$$\text{Price inc. VAT} = \text{Pre-tax price} \times \left(1 + \frac{\text{VAT\%}}{100}\right)$$

Solving GST/VAT Problems

Common problem types:

- Find the final price after adding GST/VAT
- Find the pre-tax price given the total
- Find the tax component from the total

Example: A laptop is advertised at \$1 320 including GST.
Find the pre-tax price and the GST amount.

$$\text{Pre-tax} = \frac{1\,320}{1.10} = \$1\,200$$

$$\text{GST} = \frac{1\,320}{11} = \$120$$

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Buy Now, Pay Later (BNPL)

What is it? A payment scheme allowing consumers to purchase goods immediately and pay in instalments.

Common providers: Afterpay, Zip Pay, Klarna, Humm

Typical structure:

- Purchase split into equal fortnightly instalments (often 4)
- No interest *if* paid on time
- Late fees and account fees may apply

Example: Buy a \$200 item via Afterpay.

$$4 \times \$50 \text{ fortnightly} = \$200 \text{ total (if paid on time)}$$

Analysing BNPL Costs

Potential costs to consider:

- **Late fees —** charged per missed payment
- **Account/monthly fees —** some providers charge regardless of use
- **Establishment fees —** charged on larger purchases

Example: A \$600 purchase via Zip Pay has a \$6/month account fee. If paid off over 3 months:

$$\text{Total cost} = \$600 + (3 \times \$6) = \$618$$

$$\text{Effective extra cost} = \$18$$

Key question: Is it cheaper to save and pay upfront, or use BNPL?

BNPL vs Paying Upfront — Summary

	Pay Upfront	**BNPL**
Interest	None	None (if on time)
Late fees	N/A	Yes
Account fees	N/A	Possible
Total cost	Lowest	May be higher
Accessibility	Need funds	Immediate access

BNPL can appear interest-free but hidden fees mean the true cost is often higher than paying upfront.

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Buying on Terms

What is it? Paying for goods via an initial deposit followed by regular repayments, often with added interest.

Key formula:

$$\text{Total Cost} = \text{Deposit} + (\text{Repayment} \times \text{Number of Repayments})$$

$$\text{Extra Cost} = \text{Total Cost} - \text{Cash Price}$$

Example: A TV costs \$1 200 cash. On terms: 10% deposit, then 24 monthly payments of \$55.

$$\begin{aligned} \text{Deposit} &= 0.10 \times \$1\,200 = \$120 \\ \text{Repayments} &= 24 \times \$55 = \$1\,320 \\ \text{Total cost} &= \$120 + \$1\,320 = \$1\,440 \\ \text{Extra cost} &= \$1\,440 - \$1\,200 = \$240 \end{aligned}$$

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Comparing Purchasing Methods — Example

A fridge costs \$1 800 cash. Three options are available:

Option A — Cash: \$1 800

Option B — BNPL (Afterpay): 4 fortnightly payments of \$450 + one \$10 late fee

$$\text{Total} = 4 \times \$450 + \$10 = \$1\,810$$

Option C — Terms: 20% deposit + 18 monthly payments of \$92

$$\text{Deposit} = 0.20 \times \$1\,800 = \$360$$
$$\text{Repayments} = 18 \times \$92 = \$1\,656$$
$$\text{Total} = \$360 + \$1\,656 = \$2\,016$$

Best option: Option A (cash) at \$1 800 — cheapest overall.

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Costs of Purchasing a Vehicle

The total cost of buying a vehicle includes more than just the purchase price.

Main costs to consider:

- Purchase price or loan repayments
- Stamp duty
- Registration
- Compulsory Third Party (CTP) insurance
- Non-compulsory third-party insurance
- Comprehensive insurance

Stamp Duty

Stamp duty is a government tax on the purchase of a vehicle, calculated as a percentage of the vehicle's value.

NSW rates (current):

- $0 – $44 999: 3% of market value
- $45 000 – $74 999: $1 350 + 5% of value over $45 000
- $75 000+: $2 850 + 5% of value over $75 000

Example: Vehicle valued at $32 000.

$$\text{Stamp duty} = 0.03 \times \$32\,000 = \$960$$

Registration & Insurance

Registration: Annual fee paid to the government to legally drive the vehicle. Varies by state, vehicle type, and weight.

CTP Insurance (Green Slip):

- *Compulsory* in all states
- Covers injury to other people in an accident
- Must be held to register a vehicle

Comprehensive Insurance:

- *Optional* but strongly recommended
- Covers damage to your vehicle and others' property
- Annual premiums vary by vehicle, age, and location

Total Vehicle Cost — Example

A car is purchased for \$28 000. Calculate the first-year total cost.

Cost Item	**Amount**
Purchase price	\$28 000.00
Stamp duty (3%)	\$840.00
Registration	\$380.00
CTP insurance (Green Slip)	\$550.00
Comprehensive insurance	\$1 200.00
Total first-year cost	**\$30 970.00**

Loan Repayments

If the vehicle is purchased with a loan, repayments replace the upfront purchase price.

Total loan cost formula:

$$\text{Total Loan Cost} = \text{Repayment} \times \text{Number of Repayments}$$

$$\text{Total Interest Paid} = \text{Total Loan Cost} - \text{Principal}$$

Example: \$20 000 loan, 48 monthly repayments of \$480.

$$\text{Total paid} = 48 \times \$480 = \$23\,040$$

$$\text{Interest paid} = \$23\,040 - \$20\,000 = \$3\,040$$

Add registration, CTP, and insurance on top for the true total cost.

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Homework

Complete all questions from *Managing Money — Purchasing Goods*.